Lesson 11 - Introduction to Robot Dynamics

Physical AI | Module 3: Dynamics & Control | Accelerated manual | Completed 2026-10-19

Learning outcomes

Distinguish kinematics from dynamics; interpret force, torque and rotational inertia; calculate payload gravity torque; explain the changing moment arm; verify the model in Python.

1. Kinematics versus dynamics

Kinematics describes how joint motion maps to robot positions and velocities (FK, IK, Jacobians). Dynamics describes how mass, inertia, forces and torques produce motion, which is essential for actuator selection and control.

2. Essential equations

Quantity	Equation	Meaning / units
Linear motion	F_net = m a	Force N, mass kg, acceleration m/s^2
Fixed-axis rotation	tau_net = I alpha	Torque N*m, inertia kg*m^2
Payload weight	W = m g	Weight N, g ~ 9.81 m/s^2
Gravity torque	tau_g = m g L cos(theta)	theta from horizontal; sign by convention
Horizontal / vertical	|tau_g| = m g L / 0	Max / zero magnitude for this model

Model assumptions: massless rigid arm, point payload at distance L, planar motion in a vertical plane, fixed pivot, negligible friction; balancing motor torque opposes gravitational torque. Real actuators also experience arm self-weight, inertia, friction and other loads.

3. Worked example

For m = 2 kg, L = 0.5 m, g = 9.81 m/s^2: m g L = 9.81 N*m. Gravity changes its moment arm as the arm rotates, not the payload mass.

Arm angle	Gravity torque magnitude
0 deg (horizontal)	9.810 N*m
30 deg	8.496 N*m
60 deg	4.905 N*m
90 deg (vertical)	0.000 N*m

4. Python experiment

Run with NumPy and Matplotlib; save as lesson_11_gravity_torque.py.

```
import numpy as np
import matplotlib.pyplot as plt

mass, length, gravity = 2.0, 0.5, 9.81
```

```
angles_deg = np.linspace(0, 90, 181)
torque = mass * gravity * length * np.cos(np.deg2rad(angles_deg))

for angle in [0, 30, 60, 90]:
    value = mass * gravity * length * np.cos(np.deg2rad(angle))
    print(f"Angle {angle:2d} deg: {value:.3f} N*m")

assert np.isclose(torque[0], 9.81)
assert np.isclose(torque[-1], 0.0, atol=1e-10)
print("Verification: PASSED")

plt.plot(angles_deg, torque)
plt.xlabel("Arm angle from horizontal (degrees)")
plt.ylabel("Gravity torque magnitude (N*m)")
plt.title("Lesson 11: Payload Gravity Torque")
plt.grid(True)
plt.tight_layout()
plt.savefig("lesson_11_gravity_torque.png", dpi=160)
plt.show()
```

5. Isaac Sim connection

The earlier two-link IK stage contains visual cubes with authored transforms. It does not yet implement physics-driven articulation or gravity-compensated control. Later lessons will add dynamics and actuators.

6. Assessment and outcomes

Reported Python result: PASSED. In the initial assessment, both conceptual answers were incorrect. After targeted reinforcement, the user correctly identified the shrinking gravitational moment arm and the role of moment of inertia. Lesson status: Completed. Local screenshots and source files were not independently inspected.

7. Review

Weight m*g remains constant with arm orientation. Gravitational torque reaches zero at vertical because the moment arm is zero. Increasing I increases torque required for a fixed angular acceleration. Zero gravitational holding torque does not imply zero torque during dynamic motion.

Next: Lesson 12 - Robot Equations of Motion

Combine inertia, damping, gravity and actuator torque in a single-joint equation; simulate it numerically.